

Kirubaraj Manimathavan

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EXPERIENCE

Amazon

January 2025 – Present

Software Development Engineer II – Applied AI

- Early engineer on the **Agentic AI Platform** powering **1M+** AI agents/day across Amazon, saving **\$950M+** annually.
- Built a framework-agnostic tool integration layer enabling direct API invocation as a context-efficient alternative to MCP, doubling agent tool call capacity and enabling customers to compose tools into reusable **agent skills**.
- Architected a 3-stage **AI evaluation** pipeline to maintain agent tool selection accuracy as the registry scales to **10K+** tools — integrating tool definition quality, semantic uniqueness, and shortlister regression tests as deployment gates.
- Led design, development of extensible Event-based Triggers SDK, replacing manual workflows with self-service automation — powering **30%** of all platform executions at **25K+** events/second, later extended for tool-level triggering.
- Built an AI-native developer workflow with a spec-driven **coding agent** and 6 core skills that got adopted & extended to 13 agents/19 skills across **80 engineers** driving **89%** agent adoption and **25%** faster design-to-implementation.

Troutwood

January 2024 – December 2024

AI Software Engineer – Contract

- Led 4-member team to build multi-step LangChain LLM agents reducing research time of financial advisors by **50%**.
- Implemented a persistent **agent memory** layer with multimodal RAG pipeline backed by multi-tenant Weaviate vector stores on AWS EKS, reducing agent tool calls by **30%+** and ensuring enterprise data privacy and scalability.
- Developed API-based Tools and parallel tool calling for LangChain Bedrock, increasing chat performance by **8X**.

Oracle

November 2022 – July 2023

Senior Software Engineer

- Led a cross-functional team of 3 in technical design, development, and deployment (CI/CD pipeline) of multiple full-stack enterprise software projects for Oracle ERP, resulting in successful onboarding of **3** enterprise customers.
- Spearheaded architecture design and end-to-end development of search and auto-fill features using Elasticsearch with real-time data ingestion and monitoring with Logstash and Kibana, boosting search result click rate by **75%**.
- Transformed Monolith/JSF architecture of Flexfield component to independently deployable React component with backend Node.js microservices deployed on Kubernetes, enabling portability and scalability across **12 teams**.

Software Engineer

August 2020 – November 2022

- Unified multiple custom PL/SQL packages and data models into seeded parametrized offerings avoiding code redundancy and increasing maintainability, resulting in decrease of inbound bugs by **27%** in the reporting product.
- Reconstructed the report visualization engine by parallelizing SOA service calls, redesigning React renderer and tuning the underlying SQL queries by optimizing their joins, indexes and query hints reducing rendering time by **70%**.

Eat From Home (TOVO Technologies Private Limited) [\[GitHub\]](#)

August 2018 – August 2020

Co-Founder and CTO

- Designed and developed an Android mobile app for on-demand e-commerce with **45,000+** installs and **3,000+** DAUs.
- Built and managed a Node.js REST API server for integration of user, rider and chef apps using Shopify GraphQL APIs, Razorpay webhooks, Firebase database for efficient order management ensuring real-time order tracking.
- Optimized order allocation system through order batching using Kafka and Samza reducing delivery cost by **35%**.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Software Engineering

December 2024

Coursework - Deep Learning, Cloud Computing, Distributed Systems, Data Structures, Algorithms, Software Architecture

SKILLS

AI/ML – Python | Multi-Agent Orchestration, Coding Agents, Agent Skills, Prompt Engineering, RAG, LLM Evaluation
Platform/Infra – PyTorch, LangChain, Bedrock AgentCore, Agent Memory, Claude Code, Cursor | AWS, Kubernetes
Backend/Data – Java, NodeJS, MCP, Tool Use | DynamoDB, MongoDB, Vector DB, Redis, Elasticsearch, Kafka
Full Stack – Typescript, ReactJS, NextJS | REST API, Web Sockets, SSE, gRPC, GraphQL

PROJECTS

Integrating Short-Term News Sentiment To Stock Prediction [\[GitHub\]](#), [Thesis](#), [Demo](#)

January 2024 – May 2024

- A short-term stock prediction model using attention techniques to integrate news sentiment to historical stock data.
- Implemented **3** variations of Transformer architecture on baseline CNN-LSTM model, improving MSE loss by **94.8%**.

Social Network Page Ranking using Iterative Processing on Spark

September 2023 – October 2023

- Developed Page Rank algorithm in Scala, with RDD caching and optimized network shuffles reducing runtime by **67%**.